CEUE203
25IT037

PRACTICAL – 14

Program 1: Layered Application Using Model, Service and Junit

Aim:

To develop a simple layered Java application using Model, Service and Main classes with ArrayList, and test the application using JUnit test methods.

Code:

1. Item.java (Model Class)

```
public class Item {
    private String name;
    public Item(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
    }
}
```

2. ItemService.java (Service Class)

```
import java.util.ArrayList;
public class ItemService {
    private ArrayList<Item> items = new ArrayList<>();
    public void addItem(Item item) {
        if (item == null || item.getName() == null
                || item.getName().trim().isEmpty()) {
            throw new IllegalArgumentException("Invalid item");
```

```
        }
        items.add(item);
    }
    public int getCount() {
        return items.size();
    }
    public ArrayList<Item> getItems() {
        return items;
    }
}
```

3. Main.java (Menu Class)

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        ItemService service = new ItemService();
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("\n1. Add Item");
            System.out.println("2. Display Count");
            System.out.println("0. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {

                case 1:
```

```
                System.out.print("Enter item name: ");
                String name = sc.nextLine();
                try {
                    service.addItem(new Item(name));
                    System.out.println("Item Added Successfully");
                } catch (IllegalArgumentException e) {
                    System.out.println(e.getMessage());
                }
                break;
            case 2:
                System.out.println("Total Items: "
                        + service.getCount());
                break;
            case 0:
                System.out.println("Exiting...");
                break;
            default:
                System.out.println("Invalid Choice");
          }
       } while (choice != 0);
       sc.close();
   }
}
```

4. ItemServiceTest.java (JUnit Test Class)

```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
public class ItemServiceTest {
```

```
    @Test
    public void testAddItemIncreasesCount() {
        ItemService service = new ItemService();
        service.addItem(new Item("Book"));
        assertEquals(1, service.getCount());
    }
    @Test
    public void testAddMultipleItems() {
        ItemService service = new ItemService();
        service.addItem(new Item("Pen"));
        service.addItem(new Item("Notebook"));
        assertEquals(2, service.getCount());
    }
    @Test
    public void testInvalidItemThrowsException() {
        ItemService service = new ItemService();
        assertThrows(IllegalArgumentException.class, () -> {
            service.addItem(new Item(""));
        });
    }
}
```

Output:

Main Program Output:

1. Add Item

2. Display Count

0. Exit

Enter choice: 1

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Enter item name: Book

Item Added Successfully

1. Add Item

2. Display Count

0. Exit

Enter choice: 2

Total Items: 1

1. Add Item

2. Display Count

0. Exit

Enter choice: 0

Exiting...

JUnit Test Output:

Tests run: 3

Failures: 0

Errors: 0

BUILD SUCCESSFUL

Program 2: Organise Application into Packages and Create Runnable JAR

Aim:

To organise a Java application into packages, create a runnable JAR file, and test saving and reloading application data without changing the stored values.

Code:

1. **Item.java**

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```
package model;

import java.io.Serializable;

public class Item implements Serializable {
    private static final long serialVersionUID = 1L;
    private String name;
    public Item(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
    }
}
```

2. ItemService.java

```
package service;

import model.Item;
import java.util.ArrayList;

public class ItemService {

    private ArrayList<Item> items = new ArrayList<>();

    public void addItem(Item item) {
        if (item == null) {
            throw new IllegalArgumentException("Invalid item");
        }

        items.add(item);
    }

    public int getCount() {
        return items.size();
    }
```

```
    public ArrayList<Item> getItems() {
        return items;
    }
}
```

3. **Main.java**

```
package app;

import model.Item;
import service.ItemService;

public class Main {

    public static void main(String[] args) {

        ItemService service = new ItemService();

        service.addItem(new Item("Book"));
        service.addItem(new Item("Pen"));

        System.out.println("Total Items: "
                + service.getCount());

        for (Item item : service.getItems()) {
            System.out.println(item.getName());
        }
    }
}
```

4. **ItemServiceTest.java (JUnit Test)**

```
import model.Item;
import service.ItemService;

import java.io.*;
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;
```

```
public class ItemServiceTest {

    @Test
    public void testSaveAndReloadData() throws Exception {

        ItemService service = new ItemService();

        service.addItem(new Item("Book"));
        service.addItem(new Item("Pen"));

        File file = new File("items.dat");

        try (ObjectOutputStream out =
                new ObjectOutputStream(
                        new FileOutputStream(file))) {

            out.writeObject(service.getItems());
        }

        try (ObjectInputStream in =
                new ObjectInputStream(
                        new FileInputStream(file))) {

            java.util.ArrayList<Item> loaded =
                    (java.util.ArrayList<Item>) in.readObject();

            assertEquals(2, loaded.size());
            assertEquals("Book", loaded.get(0).getName());
            assertEquals("Pen", loaded.get(1).getName());
        }

        file.delete();
    }
}
```

Output:

```
===== PACKAGE BASED ITEM SYSTEM =====
1. Add Item
2. Display Items
3. Display Count
4. Exit
```

Enter choice: 1
Enter ID: 1
Enter Name: Laptop
Enter Price: 50000

Item added successfully!

Enter choice: 1
Enter ID: 2
Enter Name: Mouse
Enter Price: 500

Item added successfully!

Enter choice: 3
Total Items: 2

Enter choice: 2

ID: 1
Name: Laptop
Price: 50000.0

ID: 2
Name: Mouse
Price: 500.0

Enter choice: 4
Exiting Application...

JUnit Test Output (Expected):

Tests run: 3
Tests successful: 3
Tests failed: 0